Yu-Sheng (Kevin) Li

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INDUSTRY EXPERIENCES

Waymo, Taiwan — Dec 2024 – present

- ML tooling Software engineer
  - Building better software stack for ML models, including profiling and simulation tools

Transperfect — Jul 2025 – present

- Part-time subtitle translator
  - 99.9 Criminal Lawyer: Seasons 1-2

Google, Taiwan — Jul 2024 – Dec 2024

- Software engineer, Google Cloud Platforms
  - Deploy and maintain cloud GPU system software

Dcard, Taiwan — May 2021 – Jun 2024

- Machine learning engineer, ads
  - *Main goal: Maximizing ads revenue (with advertiser constraints) by optimizing online ads recommender systems*
  - Implement automatic weekly model re-training and deployment
  - Simplify ranking logic by removing 3 hard rules
    → CTR +12%
  - Survey and align offline evaluation metrics with online metrics
    → from MAP to gAUC and normalized entropy (scaled log loss)
  - Mentor and help onboard another MLE
  - Design a new A/B testing architecture for cost-per-click (CPC) bidding
    → from member-based to impression-based testing
    → from member bootstrapping to member bucketing with proportional budget split
  - Design and implement the CTR calibration framework for CPC ads
    → CPC goal achieve rate from 0% to 40% and then 80%
  - Design and implement an automatic price adjustment mechanism for CPC ads
    → budget-loss adsets rate drop by 40-60%
  - Improve LLM prompt engineering and design evaluation metrics for automatic ad title revision
    → adoption rate of revised version +50%
  - Propose a hierarchical ranking framework with first ad-set ranking and second creative bandit selection
    → automatic traffic allocation to creatives with better performance
  - Feature engineering on contextual information (time, forum, position), ad industries, text-based tags, and cross features
  - Create a trouble-shooting dashboard for non-engineers
- Machine learning engineer, social
  - *Main goal: Maximizing user engagement by personalizing homepage post feed*
  - Rule-based hate speech traffic control
    → 10x reduction of daily hate speech comments

- Remove hard rules and implement embedding-based candidate for post topic recommender system
  $\rightarrow$ overall topic selection CTR +8% and overall topics per post +14%
- Improve the training set sampling strategy
  $\rightarrow$ homepage duration +0.7% and 10x distinct members in dataset (bias reduced)
- Implement item-CF (EASE^R) recall candidate and replace three rule-based candidate
  $\rightarrow$ homepage PV +1.6%
- Propose a new offline evaluation metric named normalized mean average precision (nMAP) with similar spirit of nDCG
  $\rightarrow$ aligns more consistently with online metrics

Google, Taiwan — Jul 2019 – Sep 2019

- Software engineering intern
  - Identification and interactive clustering of test failure patterns for audio components

Alibaba Group, China — Jul 2017 – Sep 2017

- Software engineering intern
  - Training time reduction by 20%-30% of distributed training on parameter servers (which motivates Publication [3])

ACADEMIC EXPERIENCES

RIKEN & Tohoku University NLP Lab, Japan — Sep 2019 – Jan 2020

- Exchange research program (advisor: Prof. Kentaro Inui)
  - Context-aware revision for academic writing assistance

Teaching Assistant, National Taiwan University

- Formal languages and automata theory — 2017 Fall, 2018 Fall, 2020 Fall
- Data structures and algorithms — 2017 Spring, 2020 Spring

KDD Cup 2017, NTU Team — Feb 2017 – Jun 2017

- Two tasks with final rankings **19/368** and **24/346**
- Contributions: Feature engineering, data analysis, and sequence to sequence learning

Academia Sinica, Taiwan — Jul 2016 – Sep 2016

- Research intern (advisor: Dr. Wei-Yun Ma)
  - Question answering game with knowledge bases and online resources

VOLUNTEER EXPERIENCES

Python Conference (PyCon) Taiwan

- Sponsorship team co-lead — Oct 2022 – Sep 2023
- Sponsorship team member — Mar 2022 – present

Taiwan Linguistics Olympiad Camp — Jul 2019

- Lecturer on computational thinking

Traditional Chinese translation of the Python Documentation — Aug 2023 – present

- Translator of parts of the Python 3.12 documents

PUBLICATIONS

[1] Bowen Yuan, **Yu-Sheng Li**, Pengrui Quan, Chih-Jen Lin. Efficient Optimization Methods for Extreme Similarity Learning with Nonlinear Embeddings. In *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 2021*.

[2] **Yu-Sheng Li**, Wei-Lin Chiang, Ching-pei Lee. Manifold Identification for Ultimately Communication-Efficient Distributed Optimization. In *International Conference on Machine Learning (ICML), 2020.*

[3] Wei-Lin Chiang, **Yu-Sheng Li**, Ching-pei Lee, Chih-Jen Lin. Limited-memory Common-directions Method for Distributed L1-regularized Linear Classification. In *SIAM International Conference on Data Mining (SDM), 2018.*

[4] **Yu-Sheng Li**, Chien-Hui Tseng, Chian-Yun Huang, Wei-Yun Ma. Guess What: A Question Answering Game via On-demand Knowledge Validation. In *International Joint Conference on Natural Language Processing (IJCNLP), 2017.*

EDUCATION

National Taiwan University, Taiwan

- M.S. in Computer Science (advisor: Prof. Chih-Jen Lin) Sep 2018 – Jan 2021
  - Research and master thesis: Publication [1]
- B.S. in Computer Science, with minor in Mathematics Sep 2014 – Jun 2018
  - Undergraduate GPA: 4.08 / 4.3

SKILLS SETS

Programming Languages

Python, SQL, C/C++, JavaScript, Shell scripting

Skills and tools

- Data science and machine learning: scikit-learn, xgboost, PyTorch, TensorFlow, numpy, pandas, Jupyter Notebook
- Natural language processing: gensim, nltk, fairseq, spaCy, transformers
- Workflow orchestration and ETL pipelines: Apache Airflow
- Database and feature stores: BigQuery, PostgreSQL, MongoDB
- Statistics: hypothesis testing and online A/B testing
- Knowledge in large-scale optimization

COMMUNICATION

- Mandarin (native)
- English (fluent)
- Japanese (conversational)

HOBBIES

- Exercises (basketball, jogging, squash, bouldering, gym)
- Chess
- Puzzles